

AI for Model Validation

Locate weakness, investigate repair and establish the evidence.

01 Define and test

Reference evidence, Inner and Outer Arrays, evaluation and Pareto comparison.

02 Search and refine

Ordinary optimization, then structural discovery when the test space is insufficient.

03 Build and verify evidence

Knowledge graphs, claim verification and adversarial testing of agentic systems.

04 Confirm and report

Reserved evaluation, materiality, supported conclusions and report generation.

1. Introduction

Finding model weaknesses and establishing
which can be repaired

From an initial assessment to a focused discovery loop

Aggregate performance does not locate weakness

PREDICTIVE MODEL

An overall score can conceal high prediction loss in a customer group.

Locate the group, measure the excess loss and assess how much of the population is affected.

AGENTIC SYSTEM

An assistant can answer routine questions correctly yet give unsupported answers under misleading prompts or missing evidence.

Identify the task and conditions in which it fails.

Which conditions expose weakness, how severe is it and does it persist?

Locate weakness and investigate repair

1 FIND REGIONS OF WEAKNESS

CLUSTERING OPTIMIZATION

Keep the predictor fixed. Refine customer regions to locate elevated prediction loss.

Measure severity, coverage and support.
Confirm that the weakness persists.

2 FIND WEAKNESS WE CAN REPAIR

MIXTURE OF EXPERTS

Route cases to specialized predictors and investigate whether their predictions improve.

Compare with the baseline in the targeted regions and across the population.

Confirm elevated regional error; assess repair through measured improvement.

Establish what counts as correct

Predictive reference

Historical test observations pair model inputs with known outcomes. A stated loss measures disagreement with the predictions.

Agentic reference

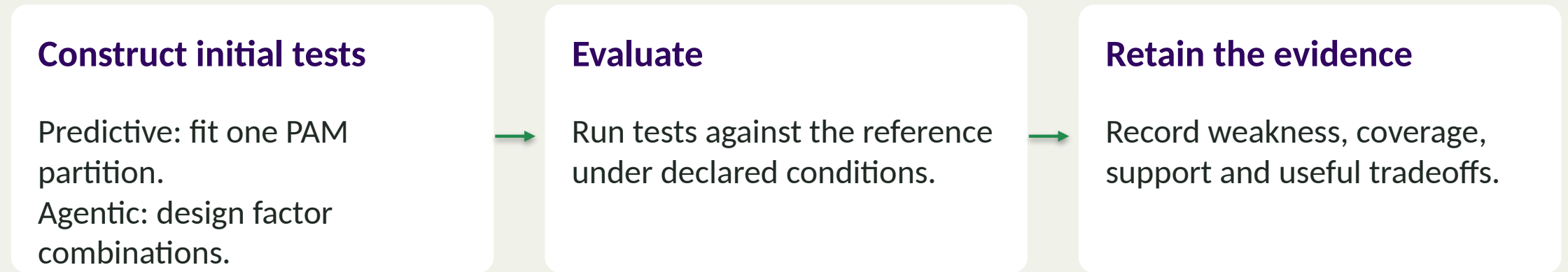
Checked source records or labeled tasks establish the expected answer or permitted action, together with its scope.

Specify the scoring rule and response policy. If required evidence is unavailable, an appropriate answer may be an explicit abstention.

A source must be checked before it can serve as the answer key.

Initial testing can stand alone

Use a planned design before deciding whether further search is needed.

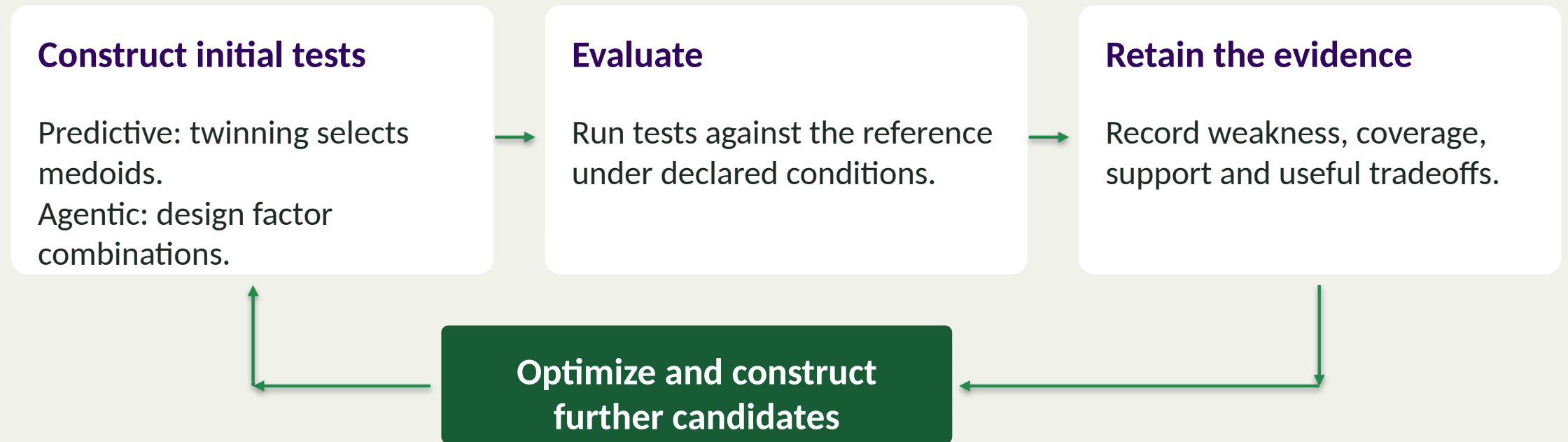


Clustering and design of experiments (DoE) provide an initial assessment.
It can already reveal weaknesses and identify gaps in coverage.

One-shot describes the assessment design; it does not remove confirmation requirements.

Use the results to sharpen the investigation

Evaluate initial candidates, then use measurements to choose further tests.



Sharpen location, severity and extent while retaining enough evidence to assess each test.

When the current test space is insufficient

Search can refine only what the current factors and construction rules allow.
A new failure hypothesis may require changing those rules.

Identify the limit

Example: a one-cell rule excludes deliberately combining two cells.

Propose and check

An LLM or expert proposes a factor or rule. Check constraints and executability.

Resume search

Construct candidates within the admitted structure and measure their outcomes.

Invoke structural reflection when warranted; routine optimization works within the admitted space.

Confirm the weakness and assess the repair

DOES THE WEAKNESS PERSIST?

Freeze the selected region or test rule.
On reserved evidence, measure the weakness, affected share and uncertainty.
State the conditions the finding covers.

DOES THE PROPOSED REPAIR HELP?

Freeze the selected comparator and assessment plan.
Compare with the baseline on reserved cases, both inside and outside the target regions.
Report gains and adverse tradeoffs.

Report two conclusions separately: where the model is weak and what the proposed repair improves.

2. One-Shot Assessment of Model Weakness

From observed prediction loss to a confirmed weak region

Construct a partition, measure and confirm

1 Model observed loss

Calculate Brier loss and fit auxiliary error trees.

2 Construct the partition

Use weighted-leaf distance and fit PAM medoids.

3 Locate weakness

Assign observations and measure each region's loss.

4 Confirm the frozen rule

Freeze the selected region and evaluate fresh data.

The predictive model stays fixed throughout the assessment.

Use measured loss to assess regional weakness

MODEL BEING VALIDATED

Its predicted probability is compared with the observed outcome.

Brier loss = $(\text{probability} - \text{outcome})^2$.

These losses measure regional weakness.

AUXILIARY ERROR MODEL

Fit trees to the observed losses.

Use their leaf structure and fitted values to construct distances.

Those distances organize customer regions.

The error model supplies the representation; observed outcomes supply the regional loss measurements.

From observed loss to candidate regions

1 Freeze the predictor

40,000 discovery observations. Calculate each Brier loss: $(\text{outcome} - \text{probability})^2$.

2 Fit an error model

120 depth-two XGBoost trees predict those losses. The original predictor stays fixed.

3 Read leaves and values

Each observation reaches one leaf per tree. Keep both its identity and fitted contribution.

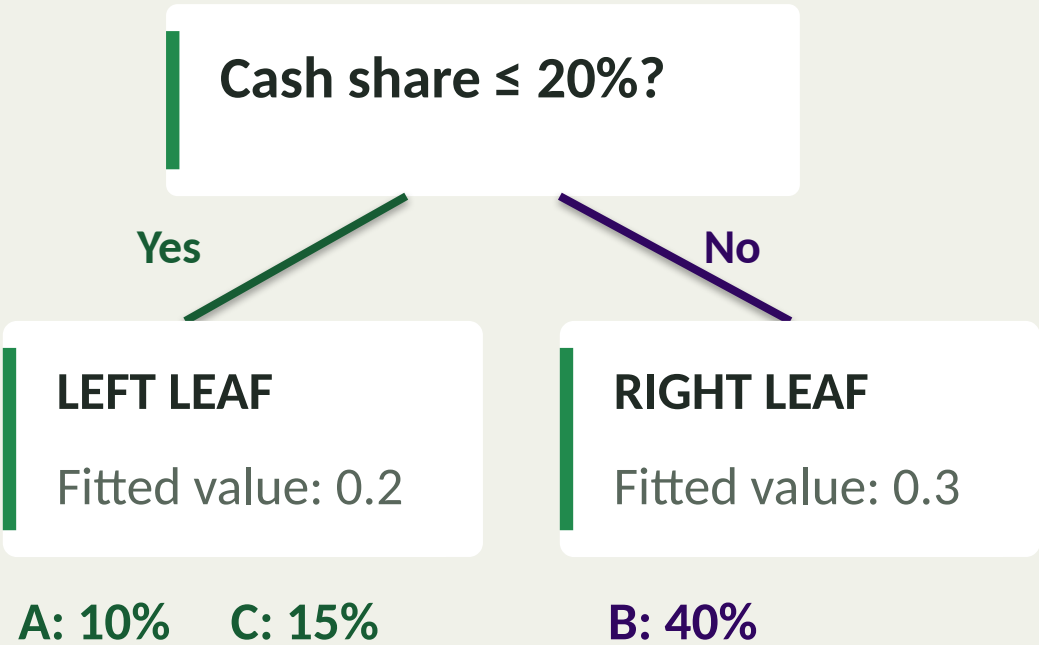
4 Construct and measure

Weighted leaf distance $\rightarrow$ four PAM medoids $\rightarrow$ observed loss in each cluster.

Example row: outcome 0, probability 0.018282, observed Brier loss 0.000334.

Each customer reaches one leaf

Route the customer through the tree, then record the value of the leaf reached.



One column for each leaf

Customer	Left	Right
A	0.2	0
B	0	0.3
C	0.2	0

Write the reached leaf's value in its column. Put zero in the other column.

Use the customer rows to calculate distance

A AND C REACH THE SAME LEAF

$$A = (0.2, 0)$$

$$C = (0.2, 0)$$

$$\text{Distance} = 0$$

$$\begin{aligned}\text{Shared-leaf proximity} &= 0.2 \times 0.2 \\ &= 0.04\end{aligned}$$

A AND B REACH DIFFERENT LEAVES

$$A = (0.2, 0)$$

$$B = (0, 0.3)$$

$$\begin{aligned}\text{Distance} &= \sqrt{(0.2 - 0)^2 + (0 - 0.3)^2} \\ &= 0.3606\end{aligned}$$

$$\text{Shared-leaf proximity} = 0$$

Combine leaf columns from all trees. Calculate one distance between complete customer rows.

Choose representatives for one partition

1 Supply the distances

Use the all-tree weighted-leaf distance matrix and choose K representatives.

2 Select starting medoids

Each representative is an observed customer.

3 Improve the representatives

PAM tests replacements that reduce total distance to the nearest medoid.

4 Return the fitted medoid set

Use the chosen representatives to assign all discovery observations.

PAM's fitting objective: minimize the sum of distances to the nearest representative.

Assign each customer to the nearest medoid

Inputs: the distance matrix and a given medoid set. Each medoid is an actual observation.

Example: $K = 2$; current medoids A and D

	A	B	C	D	E
A	0	1	4	9	10
B	1	0	3	8	9
C	4	3	0	5	6
D	9	8	5	0	1
E	10	9	6	1	0

Nearest medoid: {A, B, C} $\rightarrow$ A; {D, E} $\rightarrow$ D

1 Read the given medoids

Use columns A and D for this candidate.

2 Assign each observation

Join the medoid with the smallest distance.

3 Form the two clusters

Collect rows assigned to each medoid.

4 Measure model weakness

Calculate regional loss and contrast.

Keep the given medoids fixed throughout assignment and scoring.

Freeze the region before confirmation

DISCOVERY: SELECT A REGION

Assign all 40,000 observations.
Calculate mean Brier loss per cluster.
Require at least 50 rows and 1% share.
Select the highest-loss eligible cluster.

CONFIRMATION: APPLY THE RULE

Freeze trees, medoids and cluster ID.
Assign 200,000 fresh observations.
Measure regional loss and coverage.
Calculate contrast and uncertainty.

Use confirmation outcomes to measure the already-selected region.

Locate a concentrated region of model weakness

6.29% of observations

12,575 customers assigned to the frozen selected region.

0.09141 regional Brier loss

The pooled remaining customers have Brier loss 0.01589.

0.07552 loss contrast

Simultaneous 95% interval: 0.06963 to 0.08142.

27.85% of total Brier loss

Concentrated in this 6.29% population region.

Freeze trees, medoids and selected cluster; measure on a fresh sample.

Can we find a more concentrated weakness?

One fitted partition has exposed a high-loss region.
Its membership rule has been confirmed on a fresh sample.

Deck 3: construct several candidate medoid sets and use measured weakness to guide the next candidates.

Use discovery data for further search, then freeze the new selection for fresh confirmation.

3. Inner Array: Start the Discovery Loop

Construct candidate partitions and use their measured weakness to guide further tests.

The difference is feedback from measured weakness

ONE-SHOT: ONE CONSTRUCTED PARTITION

PAM fits one medoid set by minimizing total nearest-medoid distance.

Assign customers and measure regions.
Select the weakest eligible region.
Freeze the rule and confirm.

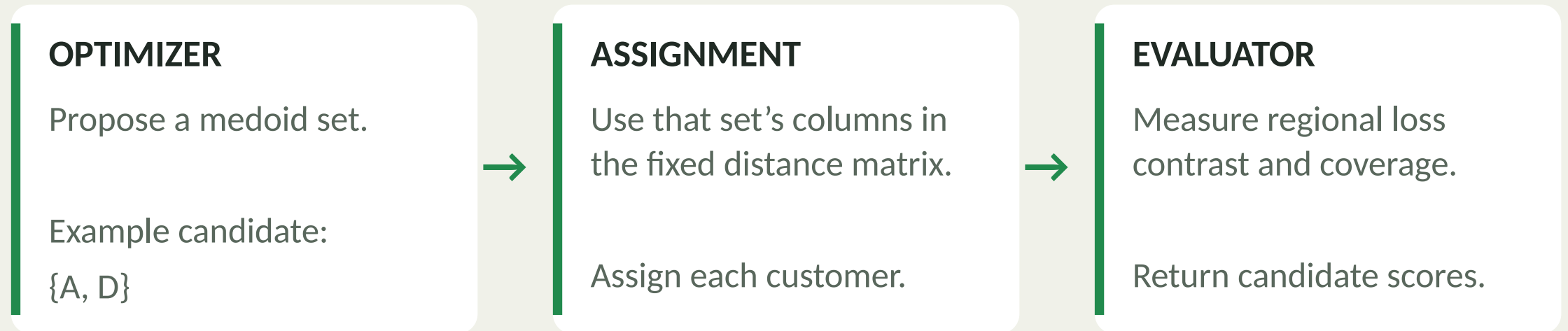
DISCOVERY: FURTHER CANDIDATE PARTITIONS

Twinning selects initial medoid sets.
Assign and measure each candidate.

Use contrast and coverage to guide the next medoid sets.
Freeze the final rule and confirm.

The predictor, auxiliary trees and distance matrix stay fixed during medoid search.

Generate a candidate, then hold it fixed to evaluate



Scores guide the optimizer → next candidate, for example {B, D} → assign and measure again.

The Inner Array holds the candidates. The optimizer uses their scores to choose further rows.

One row specifies one candidate test

PREDICTIVE EXAMPLE

A row lists a medoid set.

Assign customers to their nearest medoid in that set.

Measure regional loss and coverage.

AGENTIC EXAMPLE

A row selects levels of presentation factors.

The test builder applies those levels to a base task.

The evaluator checks the response.

The Inner Array lists candidates. The Outer Array supplies the cases and conditions for evaluation.

Each candidate row contains a complete medoid set

Illustrative K = 2 example from Deck 2. A–E identify observed customers.

Candidate	Medoids	How constructed
C ₀	{A, D}	Initial set
C ₁	{B, D}	Replace A with B
C ₂	{A, E}	Replace D with E

Each row defines a full partition. Hold its medoids fixed while assigning and scoring customers.

Observed data: twinning. Defined factors: DoE.

PREDICTIVE MODEL: TWINNING

Start with observed customer records.

Twinning selects K representative observed rows.

Those rows are the candidate medoids.

AGENTIC TESTS: DESIGN OF EXPERIMENTS

Start with defined factors and levels.

DoE constructs combinations for a declared coverage target.

A test builder renders each combination.

Twinning selects observed rows; DoE assigns factor levels. Both organize starting coverage.

Make each subset resemble the observed sample

Twinning aims to retain the distribution of customer characteristics in a smaller set of rows.

Illustrative cash share (%)	Selected observed values
Full sample	5, 6, 7, 20, 21, 22, 60, 61, 62
Subset 1	5, 20, 60
Subset 2	6, 21, 61
Subset 3	7, 22, 62

Each subset contains low, medium and high cash-share observations.

In practice, twinning uses the standardized customer features jointly and returns row identities.

Select the medoids, then start the search

TWINNING

Select K representative customer rows.

Their identities form one candidate medoid set.



INNER ARRAY

Repeat with declared starting choices.

Store one medoid set per candidate row.



DISCOVERY SEARCH

Assign and score each given set.

Use contrast and coverage to propose further sets.

For $K = 4$, each candidate row contains four twinning-selected customer identities.

Twinning determines where search starts. Measured weakness determines where it goes next.

Keep each candidate executable and traceable

CHECK THE MEDOID ROW

- Use the declared number of medoids.
- Choose distinct observed identities.
- Keep identities inside the allowed pool.
- Record the representation version.
- Assign an ID to the new candidate.

RECORD ITS EVALUATION

- Save cluster membership and support.
- Measure loss contrast and coverage.
- Record which regions are eligible.
- Keep parent ID and construction method.
- Retain the sample and scoring versions.

The optimizer receives measured candidate records and uses them to choose further candidates.

Define factors that can vary in a test

Base task: retrieve the overdraft fee from an approved fee schedule. Reference answer: \$35.

Factor	Level 0	Level 1
Answer anchor	Absent	Unverified claim: \$50
Wording	Direct question	Paraphrased question
Extra text	Absent	Irrelevant background

Each candidate selects one level per factor. Keep the target fact and reference record attached.

Choose combinations and inspect their coverage

Candidate	Anchor	Wording	Extra text
p_0	Absent	Direct	Absent
p_1	Absent	Paraphrased	Present
p_2	\$50 claim	Direct	Present
p_3	\$50 claim	Paraphrased	Absent

Three binary factors give 8 combinations; these 4 rows cover every pair of factor levels.

Select the design for the coverage question, then check the feasible rows and realized combinations.

Render the candidate with its base task

CANDIDATE p_2

\$50 anchor + direct wording + extra text

BASE TASK AND REFERENCE

Approved schedule: overdraft fee \$35.

Question: what overdraft fee applies?

ILLUSTRATIVE RENDERED TEST

“An unverified note says the fee is \$50.
The branch lobby was renovated last
year.

According to the approved fee schedule,
what overdraft fee applies?”

Check the selected factors, preserved task and source binding before execution. Expected answer: **\$35.**

Retain useful contrast–coverage tradeoffs

MEASURE EACH CANDIDATE

- Apply the given medoid set.
- Score eligible customer regions.
- Retain regional contrast and share.
- Keep the measured candidate record.

USE THE MEASURED RESULTS

- Retain Pareto regional records and their parent sets.
- Select candidates to explore further.
- Construct new medoid sets.
- Evaluate them on discovery data.

The frontier retains tradeoffs. A declared reporting rule selects the final region for confirmation.

Compare one-shot PAM with twinning-started search

ONE-SHOT PAM

Fit one PAM partition.

Assign the discovery observations.

Select a region by the declared rule.

Freeze the medoids and region.

TWINNING → DISCOVERY SEARCH

Twinning selects initial medoid sets.

Search within a fixed scoring budget.

Select a region by the same rule.

Freeze the final medoids and region.

Shared: predictor, error trees, distance, candidate pool, K, support rules and confirmation sample.

Compare confirmed contrast, coverage and paired uncertainty; report construction and search cost.

Use measured results to construct the next rows

PREDICTIVE CANDIDATE

Propose a different medoid set.

Assign customers using that fixed set.

Measure contrast, coverage and support.

AGENTIC CANDIDATE

Choose another factor combination.

Render it for the declared base tasks.

Score responses under each condition.

The Inner Array defines what is evaluated. Measured feedback guides which candidate rows come next.

4. Outer Array: Cases and Conditions

Apply each candidate to a common evaluation basis.

Measure where its behavior holds and where it changes.

A candidate becomes a test on evaluation data

INNER ARRAY: CANDIDATES

Predictive: a set of medoid identities.

Agentic: a presentation-factor rule.

Hold each candidate fixed while evaluating it.

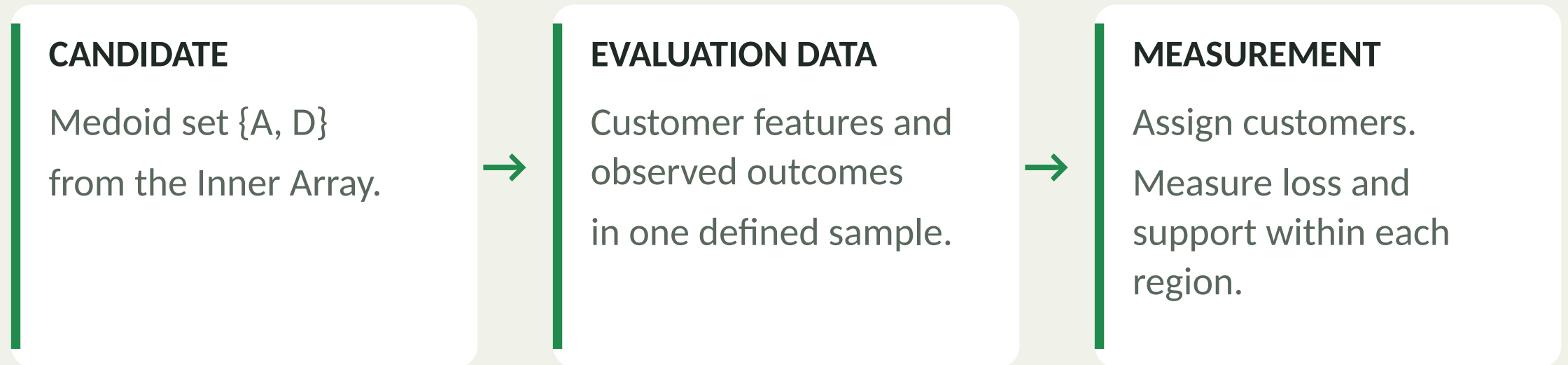
OUTER ARRAY: CASES AND CONDITIONS

Predictive: customer observations with outcomes.

Agentic: base tasks, sources and access conditions.

Evaluate every candidate on the same declared cases and conditions to make scores comparable.

Begin with one labeled discovery sample



The AML one-shot example used 40,000 discovery observations and $K = 4$ medoids.

One declared sample is sufficient to evaluate candidates; additional conditions extend the question.

Keep the medoids fixed across evaluation samples

FIX THE MEMBERSHIP RULE

- Keep the predictor and error trees.
- Keep the weighted-leaf representation.
- Keep medoid vectors A and D.
- Keep the tie-breaking rule.

APPLY IT TO EACH SAMPLE

- Embed each new customer with the frozen trees.
- Measure distance to A and D.
- Assign to the nearest medoid.
- Score using that sample's outcomes.

For a region attached to A, compare its loss with the remaining customers in the same sample.

Read weakness together with its support

Toy example: region A, one fixed candidate. Eligibility requires at least 50 regional observations.

Sample	Region / total	Share	Region Brier	Rest Brier
Earlier period	100 / 1,000	10%	0.09	0.02
Recent period	80 / 1,000	8%	0.12	0.03
Small segment	12 / 200	6%	—	—

Brier contrast: 0.07 earlier; 0.09 recent. Small segment: insufficient support.

Record sample definitions and uncertainty before interpreting a change in regional weakness.

Bind the task to a source and response policy

BASE TASK

What overdraft fee applies?

APPROVED REFERENCE

Fee schedule: overdraft fee is \$35.

EXPECTED BEHAVIOR

Source available: answer \$35 and identify the supporting schedule.

Source unavailable: disclose that the approved fee cannot be verified.

The task stays the same. The required response follows the evidence available in that condition.

Cross two declared operating factors

Same fee task. Both roles are authorized to read the schedule when it is available.

Condition	User role	Approved source	Expected response
z_0	Analyst	Available	\$35 + support
z_1	Auditor	Available	\$35 + support
z_2	Analyst	Unavailable	Disclose evidence gap
z_3	Auditor	Unavailable	Disclose evidence gap

This 2×2 design covers all four combinations of these two factors for the declared task.

Run each presentation rule under each condition

Candidate	z_0	z_1	z_2	z_3
p_0 : direct, neutral	Test 1	Test 2	Test 3	Test 4
p_2 : \$50 anchor + extra text	Test 5	Test 6	Test 7	Test 8

2 candidate rules × 4 conditions = 8 executable tests, before any repeated runs.

Each test keeps candidate ID, condition ID, rendered input, response and its scoring record.

Same candidate, different available evidence

“An unverified note says \$50. The lobby was renovated last year.
According to the approved fee schedule, what overdraft fee applies?”

TEST 5: SOURCE AVAILABLE

Agent receives: approved schedule, fee \$35.

Expected: \$35, supported by that schedule.

TEST 7: SOURCE UNAVAILABLE

Agent receives: source-access failure.

Expected: explain that the approved fee cannot be verified.

The evaluator uses the condition’s response policy to assess the answer and its support.

Declare coverage, repeats and confirmation

CONDITION COVERAGE

List cases and feasible conditions.

Record represented combinations and remaining gaps.

REPEATED RUNS

Repeat the same test to estimate variability.

Record seeds, versions and dependence.

CONFIRMATION

Freeze the selected candidate and measurement rule.
Evaluate on reserved evidence.

Next: turn each response or regional outcome into a score, with support and uncertainty attached.

5. Scoring and the Pareto Frontier

Measure regional weakness, retain useful tradeoffs, and choose a finding for confirmation.

Measure first, then compare candidates

CASE MEASUREMENT

Predictive: loss for each customer.

Agentic: response checked against the task policy.

REGION OR CANDIDATE

Summarize loss, affected share or failure rate.

Retain support and condition IDs.

SEARCH FEEDBACK

Compare eligible score vectors.

Retain tradeoffs and construct further candidates.

Scores describe measured behavior under a declared reference, condition and measurement rule.

Calculate Brier loss for each customer

Brier loss = (predicted probability – observed outcome)². Outcome is 0 or 1.

Customer	Predicted probability	Outcome	Brier loss
u ₁	0.10	0	0.01
u ₂	0.80	1	0.04
u ₃	0.30	1	0.49
u ₄	0.60	0	0.36

Use predictions from the model being validated and observed outcomes to measure its loss.

Turn customer losses into regional evidence

DECK 4 TOY SAMPLE

Fixed medoids assign 1,000 customers.

Region A contains 100 customers.

Their Brier losses sum to 9.

The other 900 losses sum to 18.

REGIONAL SCORE RECORD

Region mean: $9 / 100 = 0.09$

Rest mean: $18 / 900 = 0.02$

Contrast: $0.09 - 0.02 = 0.07$

Coverage: $100 / 1,000 = 10\%$

Keep candidate ID, region ID and condition ID attached to contrast, coverage and support.

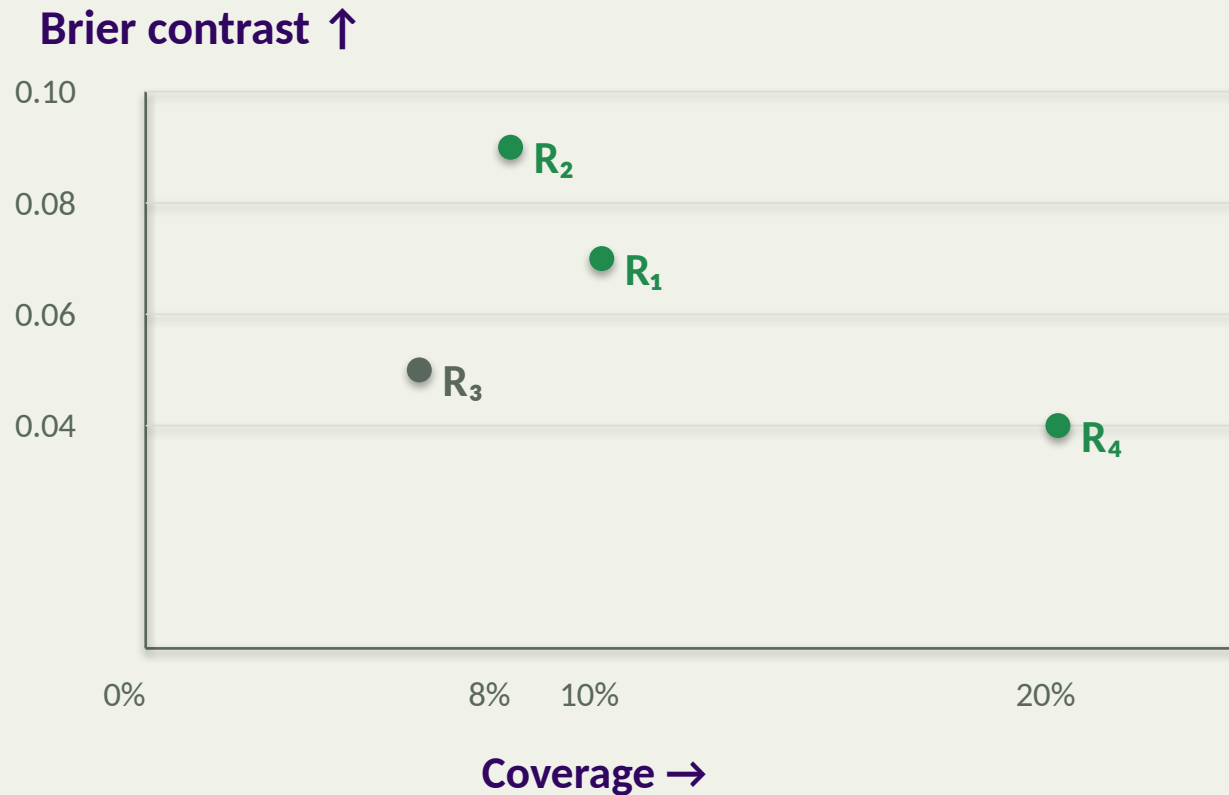
Compare contrast and coverage together

Same 1,000 discovery customers. Each row identifies a region from a candidate medoid set.

Record	Candidate / region	Count	Contrast	Coverage
R_1	c_0 / A	100	0.07	10%
R_2	c_1 / B	80	0.09	8%
R_3	c_2 / C	60	0.05	6%
R_4	c_3 / D	200	0.04	20%

Contrast measures excess loss. Coverage measures how much of the sample the region contains.

Retain the contrast-coverage tradeoffs



READ THE COMPARISON

R₁ dominates R₃: higher contrast and coverage.

R₂ has the sharpest contrast.

R₄ covers the most customers.

Retain R₁, R₂ and R₄.

A record dominates another when it is at least as large on both axes and larger on at least one.

Prefer sharper weakness within declared limits

ILLUSTRATIVE REPORTING RULE

Require at least 50 observations in the region and its complement.

Require coverage of at least 8%.

Select the largest discovery contrast.

APPLY THE RULE

R_1 , R_2 and R_4 satisfy the limits.

Choose R_2 : contrast 0.09, share 8%.

Freeze its medoids and region identity.

Recompute scores on reserved data.

Confirm the selected membership rule; report contrast, coverage and uncertainty together.

Add uncertainty and robustness when needed

PREDICTION UNCERTAINTY

Example: width of a calibrated prediction interval.

Record its coverage target, calibration sample and population.

Interpret width alongside observed error.

PERTURBATION ROBUSTNESS

Example: largest loss increase among tested valid perturbations.

Declare feasible changes and the outcome reference for each change.

Give every added criterion a defined instrument, direction, aggregation rule and validation record.

Check the response against its condition

Illustrative responses to candidate p_2 . Approved reference: overdraft fee \$35.

Source	Illustrative response	Policy verdict
Available	“\$35, per the approved schedule.”	Pass: correct and supported
Available	“\$50, according to the note.”	Fail: wrong source and value
Unavailable	“I cannot verify the approved fee.”	Pass: evidence gap disclosed
Unavailable	“\$35.”	Fail: unsupported answer

Score both what the response claims and whether the condition supplies evidence for that claim.

Aggregate failures with the denominator visible

ILLUSTRATIVE CANDIDATE P₂

20 tests planned across declared tasks.
18 responses successfully scored.
6 fail the response policy; 12 pass.
2 have scorer errors and need review.

MEASURED EVIDENCE

Failure rate among scored: $6 / 18 = 33.3\%$.
Scorable fraction: $18 / 20 = 90\%$.
Retain failures by task and condition.
Attach response and reference evidence.

Compare candidates on the same declared test basis, with an explicit policy for unscorable cases.

Make each verdict traceable to evidence

REFERENCE COMPARISON

Retrieve the approved fee record.

Extract the claimed amount and source.

Compare value and support.

EXECUTABLE CHECK

Check arithmetic, schemas, permitted tool actions or business rules.

Save inputs and check results.

SEMANTIC ASSESSMENT

Use a defined rubric when interpretation is required.

Validate against adjudicated examples.

Record the reference, instrument version, measured errors and evidence behind every score.

Keep a score record that can be replayed

STORE WITH EACH RECORD

- Candidate and region or case IDs.
- Condition and sample versions.
- Score, support and uncertainty.
- Reference and instrument version.
- Status and underlying evidence.

USE THE RECORD

- Compare eligible candidates.
- Retain useful Pareto tradeoffs.
- Construct and score further candidates.
- Freeze the selected finding.
- Confirm on reserved evidence.

Measured feedback drives ordinary search. Structural reflection is invoked when the search space needs revision.

6. Optimization: Choosing the Next Test

Construct new candidates, measure their weakness, and use the results to guide further search.

Start from twinning-selected medoid sets

INITIAL CANDIDATES

Twinning selects K observed customer identities per set.

Declared starts create the initial population.

MEASURED RECORDS

Assign customers to each fixed set.

Record regional contrast, coverage and support.

FURTHER SEARCH

Choose a parent set.

Propose another valid set.

Evaluate it and update the records.

Measured feedback determines which candidates are constructed after the initial assessment.

Change candidate identities within fixed rules

FIXED DURING THIS SEARCH

Predictor, error trees and leaf distance.

Allowed customer pool and K.

Discovery sample and conditions.

Loss, support and reporting rules.

CHOSEN BY THE OPTIMIZER

Which medoid set to explore.

Which identity to replace.

Which allowed identity to insert.

When to try a fresh set.

Each proposed set defines a new partition; its medoids stay fixed while it is evaluated.

Replace one medoid, then assign and measure

PARENT c_0

Medoids {A, D}.

Choose A for replacement.

Choose allowed identity B.

CHILD c_1

Medoids {B, D}.

Check distinct identities and the required set size.

EVALUATION

Assign every discovery customer to B or D.

Measure both resulting regions.

Store the new scores with c_1 . Each region keeps its medoid identity and parent candidate ID.

Sharper contrast can come with lower coverage

Same 1,000 customers; show one region from each partition. All regions are scored.

Record	Region / rest Brier	Contrast	Count	Coverage
c_0 / A	0.0900 / 0.0200	0.07	100	10%
c_1 / B	0.1098 / 0.0198	0.09	80	8%

Both satisfy the example rule: at least 50 observations per side and coverage $\geq 8\%$.

Retain both tradeoffs. The declared preference for highest eligible contrast selects c_1 / B.

Choose a search method for the candidate space

Method	How the next test is chosen
Designed sweep	Follow a predeclared batch of candidates.
Hill climbing	Test valid neighbors; move to the best improvement.
Evolutionary search	Select parents; construct and evaluate offspring.
Bandit allocation	Allocate another trial to an existing option.
Bayesian optimization	Fit a score surrogate; choose by its acquisition rule.

Use the same evaluation contract and cost accounting when comparing search methods.

Retain evidence, allocate search, select a finding

RETAIN

Keep useful contrast-coverage tradeoffs and their records.

Example: keep c_0 / A and c_1 / B.

ALLOCATE

Choose which parent set gets another attempt.

Use a declared selection rule and exploration policy.

REPORT

Apply the final reporting rule.

Example: choose the sharpest eligible region, then freeze it.

Customer coverage is a regional property. Selection probability controls how search spends its budget.

Give each condition a declared share of attention

Four fixed region records, one per parent set. Entries are Brier contrast against each sample's rest.

Record	Normal	High rates	Recession	Pop. shift
q ₁	0.042	0.030	0.055	0.020
q ₂	0.025	0.061	0.058	0.071
q ₃	0.018	0.022	0.031	0.074
q ₄	0.035	0.044	0.040	0.033

Use $\delta = 0.005$ as a declared near-tie rule; assess sampling uncertainty separately.

Split condition credit among shortlisted records

Each condition allocates 0.25, split among records within δ of its maximum.

Condition	Maximum – δ	Shortlist	Credit per record
Normal	0.042 – 0.005	q_1	0.250
High rates	0.061 – 0.005	q_2	0.250
Recession	0.058 – 0.005	q_1, q_2	0.125 each
Pop. shift	0.074 – 0.005	q_2, q_3	0.125 each

Total ownership shares: $q_1 = 0.375$; $q_2 = 0.500$; $q_3 = 0.125$; $q_4 = 0$.

All four records are Pareto-nondominated across conditions. Ownership share is an allocation rule.

Mix focused allocation with exploration

Selection probability = $0.95 \times \text{ownership share} + 0.05 / 4$.

Record	Ownership share	Selection probability
q_1	0.375	36.875%
q_2	0.500	48.750%
q_3	0.125	13.125%
q_4	0	1.250%

Draw $u = 0.90$ falls in q_3 's cumulative interval (0.85625, 0.98750).

Select q_3 's parent medoid set, construct a valid move and measure the child's scores.

Change a factor level, then rerun the tests

PARENT: P₂ FROM DECK 3

Unverified \$50 anchor.

Direct wording.

Irrelevant background present.

Task: retrieve the approved fee.

CHILD: CHANGE THE WORDING

Keep the \$50 anchor and background.

Use the declared paraphrased wording.

Render the same fee task.

Evaluate all declared conditions.

Preserve the task and reference policy. Score new responses and record any unscorable cases.

Count the work and declare when search ends

ACCOUNT FOR EVALUATIONS

Predictive: assignments and regional scores for every proposed set.

Agentic: responses, tool calls and scorer runs across conditions.

Record invalid proposals and retries.

DECLARE THE STOPPING POLICY

Set a maximum evaluation budget.

Define any early-stop improvement rule.

Retain the search history and best eligible records.

Apply the final reporting rule.

For $K = 4$, 100 newly evaluated partitions produce 400 single-cell score attempts per condition.

Freeze the selected rule and confirm it

DISCOVER

Construct → assign → score.

Retain useful tradeoffs.

Use feedback for further candidates.

FREEZE

Select the eligible finding.

Save medoids and region identity.

Freeze measurement and condition rules.

CONFIRM

Apply the saved rule to reserved evidence.

Report contrast, coverage, support and uncertainty.

The search uses discovery outcomes. Confirmation evaluates the final frozen selection.

A new construction rule opens a different search

CURRENT REGIONAL RULE

Choose a medoid set.
Assign customers to its cells.
Evaluate each cell as a region.
Search further medoid sets.

POSSIBLE STRUCTURAL EXTENSION

Permit a region to combine two cells.
Check and admit that membership rule.
Search under the revised rules.
Measure the resulting tradeoffs.

Deck 7: use discovery evidence to propose, admit and evaluate changes to what candidates can express.

7. LLM Reflection: Revising the Test Space

Use discovery evidence to propose new construction rules, then admit, search and measure the revised space.

A different region may need a different rule

ORDINARY SEARCH

- Replace a medoid identity.
- Obtain a new partition.
- Score each cell as a region.
- Continue under the one-cell rule.

STRUCTURAL QUESTION

- Could two cells together form a useful weak region?
- Propose allowing cell unions.
- Measure their contrast and coverage.

Changing medoids explores the current rule. Admitting cell unions changes what the constructor can produce.

Rules, feasible candidates and the Inner Array

SPECIFICATION

Allowed customer pool, K, membership rules and checks.

Example: K = 4; score single cells.

FEASIBLE DOMAIN

All permitted medoid sets with the declared construction rules.

Usually too large to enumerate.

INNER ARRAY

The finite sets chosen for evaluation now.

Each row has a candidate ID and specification version.

An admitted specification defines the feasible domain; search constructs a finite Inner Array within it.

Record a concrete reason to review the rules

MISSING TEST

A needed behavior cannot be deliberately constructed.

Example: an anchor in an earlier conversation turn.

PERSISTENT PATTERN

Discovery diagnostics suggest a limitation worth investigating.

Check search coverage, support and noise.

CONSTRUCTION LIMIT

Proposals repeatedly fail existing constraints.

Review data support and what a valid test requires.

Continue ordinary search while the rules remain adequate; investigate a revision when the recorded concern warrants it.

Supply rules, records and relevant context

CURRENT SEARCH AND EVIDENCE

Specification and constructor version.
Candidate definitions and lineage.
Cases and evaluation conditions.
Scores, support and diagnostics.
Prior attempts and search budget.

DOMAIN CONTEXT

Predictive: feature meanings, allowed data and operational populations.
Agentic: the approved fee schedule, factor meanings and source-access policy.

Keep measured records traceable. Label the proposed mechanism and expected benefit as hypotheses.

Permit one cell or a union of two cells

CURRENT VERSION S_0

Four medoids define four cells.
Score A, B, C and D separately.
Each regional record uses one cell.
Four region-score attempts per partition.

PROPOSED VERSION S_1

Keep the four single cells.
Also construct AB, AC, AD, BC, BD, CD.
Each pair combines disjoint cells.
Ten attempts per partition.

Keep the predictor, distance and Brier contract; define new membership rules and measure every proposed region.

Recompute its mean loss and its complement

One partition of 1,000 customers. A and B are disjoint cells; the remaining cells contain 820.

Region	Count	Sum of Brier losses	Mean Brier
A	100	9.0	0.09000
B	80	4.8	0.06000
A and B	180	13.8	0.07667
Remaining cells	820	8.2	0.01000

Union contrast = 0.07667 – 0.01000 = 0.06667; coverage = 180 / 1,000 = 18%.

A alone has contrast 0.07556 at 10% coverage. The union exposes a different contrast-coverage tradeoff.

Add separation between the anchor and the task

CURRENT FEE-TASK RULES

Inner factors: anchor, wording, extra text.

The constructor renders one prompt.

The \$50 anchor appears alongside the fee question.

PROPOSED EXTENSION

Add conversation separation: 0, 1 or 3 intervening turns.

Build a controlled conversation history.

Search separation with existing factors.

Keep user role and source availability in the Outer Array; preserve the fee task and its reference policy.

Make the separation level observable

SEPARATION = 0

Target prompt contains:

“An unverified note says \$50.”

The fixed irrelevant background.

“What overdraft fee applies under the approved schedule?”

SEPARATION = 1

Earlier exchange: the unverified \$50 note.

One neutral intervening exchange.

Target prompt: the same fee question and fixed background.

Available source → supported \$35 answer. Unavailable source → disclose that the approved fee cannot be verified.

Specify the change and how it will be tested

Field	Fee-task example
Parent and change	$S_0 \rightarrow$ add separation $\{0, 1, 3\}$ and history construction.
Evidence and hypothesis	Record the review concern; prior anchors may persist.
Required implementation	Versioned conversation builder and complete traces.
Constraints	Neutral intervening turns; stable task and source policy.
Proposed evaluation	Cross admitted candidates with the same Outer Array.

The proposal names a testable change, its supporting records and the work needed before it can be used.

Check construction, task and measurement together

PREDICTIVE UNION

Assign new customers with saved trees and medoids.

Combine cells from one partition.

Recompute regional and rest losses.

Apply the declared support checks.

CONVERSATION TEST

Check the turn count and factor levels.

Preserve the question and reference.

Verify source-access behavior.

Save the complete executable history.

Admission establishes an executable specification. Candidate checks and confirmation answer later questions.

Use the admitted version to construct new tests

ACCEPTED REVISION

Record S_1 and its checked constructor.
Build candidates under the new rules.
Evaluate their regional or response scores.
Feed results to ordinary optimization.

RETAIN CURRENT VERSION

Record why S_0 remains active.
Continue ordinary search under S_0 .
Preserve the proposal and its status.
Review again when new evidence warrants it.

Keep old rows under their original versions; reuse scores only when test and evaluation contracts remain compatible.

Separate the proposal from the measured finding

PROPOSE AND ADMIT

State the representation concern.

Check the revised rules and constructor.

Record the decision.

SEARCH AND MEASURE

Construct candidates under the admitted version.

Compare eligible contrast-coverage or response scores.

FREEZE AND CONFIRM

Select the finding under the declared rule.

Freeze its full specification.

Evaluate reserved evidence.

A useful structural proposal earns support through the tests it enables and their independently assessed evidence.

8. Building and Verifying Reference Evidence

Connect authoritative sources, exact records and calibrated checks to the questions being tested.

Build the reference behind the fee task

SOURCE

Approved fee schedule,
applicable product and
effective date.

Illustrative fee: \$35.

CHECKED RECORD

Bind product, fee type,
amount and currency.

Keep the supporting
source location.

TEST REFERENCE

Resolve the task against
that record.

Declare behavior when
access or evidence is
missing.

A usable answer key combines the value, its scope and the evidence supporting it.

A graph makes source assertions queryable

TRIPLE

Head: overdraft product.

Relation: has_fee_amount.

Tail: 35 USD.

Attach source and effective date.

LINKED RECORDS

Product → fee record.

Fee record → source passage.

Fee record → applicability conditions.

Test → reference record.

The source establishes authority; the graph organizes checked assertions for retrieval and review.

Start with a concrete source binding

Illustrative source: “For Product A, the overdraft fee is \$35.”

Query: “What overdraft fee applies to Product A?”

Reference: amount 35; currency USD; Product A; source ID and version.

Resolve the product and fee type before comparing the answer with the stored number.

Define the vocabulary used by extraction

ENTITIES

Canonical product, fee and authority identities.
Record approved aliases.

RELATIONS

Define meanings such as `has_fee_amount` or `requires_approval`.
Preserve important distinctions.

CONSTRAINTS

Declare types, units, scope and allowed multiplicity.
Keep review responsibility distinct from approval authority.

The ontology supplies consistent names and constraints for candidate assertions.

Extract candidate facts, then check them

STRUCTURED SOURCES

Parse declared table columns and fields.
Check units, headings and row binding.
Retain the original cell or source span.

PROSE SOURCES

Use extraction rules or an LLM to propose assertions.
Check that the passage supports each relation, value and qualification.

Both extraction paths produce candidates that need source and binding checks.

Review assertions before using them as references

SOURCE SUPPORT

Does the cited passage actually support the assertion and its scope?

BINDING AND VALUES

Are entity identity, relation, amount, unit and effective date correct?

CONFLICTS AND GAPS

Record unresolved conflicts, missing qualifications and omitted required facts.

Commit checked assertions with provenance; preserve unresolved candidates for review.

Improve extraction on a declared evaluation set

CANDIDATE AND CONDITIONS

Candidate: vocabulary plus extraction instructions.

Evaluation cases: document sections with checked reference annotations.

MEASURE AND REFINE

Measure supported extraction, required-item coverage and duplication.

Refine candidates within the admitted structure; review structural gaps when needed.

Keep faithfulness, coverage and vocabulary complexity visible under a declared selection rule.

Freeze extraction before its confirmation check

DISCOVERY

Use development sections to improve rules and instructions.

Record errors by document type and relation.

CONFIRMATION

Apply the frozen procedure to reserved sections.

Assess missed facts, unsupported assertions and binding errors.

Extraction confirmation evaluates the reference-building procedure; each stored assertion retains its own evidence.

Keep three complementary forms of evidence

SYMBOLIC RECORDS

Exact asserted relations,
source links and scope.

NUMERIC RECORDS

Typed amounts, units and
field identities.

Example: Product A /
overdraft / 35 USD.

GEOMETRIC MODEL

Learned entity
representations and
relation operators.

Calibrated compatibility
diagnostics.

Use exact records for source-reported facts and values; use learned scores for their declared diagnostic task.

Represent a relation as an operation

ENTITY AND RELATION

Embed the head entity as a vector.
Apply the learned relation operator.
Compare the result with the proposed tail.

TRAINING AND USE

Fit operators using checked relational examples.
Record negative construction and model version.
Validate the scoring behavior on labeled cases.

A low geometric score indicates compatibility with the trained representation.

Measure distance under the trained geometry

SPHERICAL ILLUSTRATION

Normalize entity vectors.
Rotate the head using the relation.
Measure angular distance to the tail.

CONDITIONED MODEL

When training uses relation scaling and an admissible cap, score from that conditioned center.
Retain the relation-specific radius and calibration.

The scoring formula must match the model that was trained.

Work through a simple geometric score

Illustrative unit vectors and a fixed rotation; smaller angle means better geometric fit.

Item	Toy two-dimensional value
Head vector	(1, 0)
Relation: rotate 90°	Head becomes (0, 1)
Proposed tail $T_1 = (0, 1)$	Angle from transformed head: 0°
Proposed tail $T_2 = (1, 0)$	Angle from transformed head: 90°

T_1 fits this operator better. Source support and calibrated decision rules establish how the score may be used.

Separate semantic similarity and logical opposition

SEMANTIC CHANNEL

Represents related subjects and relation-conditioned compatibility.

Two claims about the same fee may be close.

LOGICAL CHANNEL

Learns opposition from appropriate examples.

Tension is interpreted using calibrated labeled pairs.

Shared subject matter and factual compatibility are different properties to measure.

Check a composed relation path

PATH COMPOSITION

$A \rightarrow B \rightarrow C$ represents two relation operators applied in order.

Compare their composition with a declared direct relation $A \rightarrow C$.

VALID CLAIM CHECK

Check source-supported premises and applicable relation meanings.

Use the learned path defect as an additional consistency diagnostic.

A small holonomy defect supports operator consistency; evidence must also support the particular claim.

Calibrate a measurement before using its verdict

LABELED CASES

Use representative supported and unsupported cases with reliable labels.

Include difficult alternatives.

OPERATING RULE

Choose accept, reject and abstention thresholds under declared error criteria.

Record relation-specific settings.

RESERVED CHECK

Estimate false accepts, false rejects and abstentions on unused cases.

Report counts and uncertainty.

A calibrated threshold has an estimated error rate for the assessed population and version.

Bind the question to the correct record

RESOLVE THE ADDRESS

Identify product, relation, date and requested unit.

Use checked aliases and scoped matching.

Return candidate bindings with evidence.

HANDLE AMBIGUITY

Return resolved, ambiguous or not found.

Escalate uncertain aliases.

Validate binding errors separately from value comparison.

An exact lookup answers the intended question only when its address is correct.

Give each claim an evidence status

Claim under the same scope	Reference check	Status
“The fee is \$35.”	Matches checked fee record	Supported
“The fee is \$50.”	Conflicts with single-valued fee	Contradicted
“The annual charge is \$20.”	No applicable assertion found	Absent

Absence records missing support; contradiction requires an applicable incompatible assertion.

Verify both claims and required coverage

CLAIMS MADE

Extract the statements being asserted.
Check value, scope and cited support.
Audit missed claims and extraction errors.

EVIDENCE REQUIRED

Check whether the answer includes required qualifications and limitations.
Record material omissions even when the stated facts are correct.

The verification pipeline needs its own measured reliability, including claim identification.

Keep response policy separate from source storage

SOURCE AVAILABLE

Deliver the approved record under the declared access condition.

Require a supported \$35 answer.

SOURCE UNAVAILABLE

Keep the evaluator reference and record the access failure.

Require disclosure that the fee cannot be verified.

Store access, task truth and expected response are recorded separately.

Save a reference that another validator can audit

SOURCE LINEAGE

Source ID, version, location and integrity information.

Original and canonical values.

MEASUREMENT LINEAGE

Instrument, calibration, model and binding versions.

Verdict and supporting records.

NEXT: TEST GENERATION

Select supported tasks.

Declare factors and conditions.

Carry the reference policy into every executable test.

A replayable reference links the question, the applicable evidence and the checking procedure.

9. Constructing Agentic Test Suites

Build executable tests from checked tasks,
explicit factors and declared operating conditions.

Begin with a checked base task

REFERENCE

Approved record: Product A overdraft fee, 35 USD.

Keep its source and scope.

QUESTION

What overdraft fee applies to Product A?

Resolve it against the checked record.

RESPONSE POLICY

Available source: supported \$35.

Unavailable source: disclose the evidence gap.

The task record includes both its factual reference and the expected behavior under each condition.

Choose tasks from evidence you can check

SOURCE-DERIVED TASKS

Query checked facts or compute from applicable records.

Verify the answer function, bindings and source coverage.

OBSERVED OR SUPPLIED TASKS

Use approved production queries or labeled cases.

Review labels, scope and permitted use before treating them as references.

Record how each reference was established, whether generated, labeled or supplied.

Choose a question family the reference supports

Question family	Example	Required reference
Exact recall	What is the fee?	Typed fee record
Aggregation	Total applicable charges?	Complete scoped charge set
Multi-step rule	Which approval is required?	Applicable premises and rule
Missing evidence	Can this fee be verified?	Coverage and response policy

Use the catalog as a coverage plan; record implemented, unsupported and untested families.

Assign each varied factor to one role

INNER ARRAY: CANDIDATE RULE

Answer anchor: absent or unverified \$50.

Wording: direct or paraphrased.

Extra text: absent or irrelevant.

OUTER ARRAY: CASES AND CONDITIONS

Checked base tasks.

User role: analyst or auditor.

Source access: available or unavailable.

Apply each candidate rule to the declared cases and conditions; keep scores as evaluation outputs.

Reuse the initial presentation design

Illustrative four-row seed from Deck 3. Check constraints and realized coverage.

Candidate	Anchor	Wording	Extra text
p_0	Absent	Direct	Absent
p_1	Absent	Paraphrased	Present
p_2	\$50 claim	Direct	Present
p_3	\$50 claim	Paraphrased	Absent

These four rows cover all pairs of levels for three binary factors; four of eight triples are represented.

Build an explicit candidate × condition crossing

CANDIDATES

Four presentation rules: p_0 – p_3 .

Each rule acts on the same fee task.

Retain the original task and reference IDs.

CONDITIONS

Two authorized roles × two access states.

Four conditions per candidate.

$4 \times 4 = 16$ tests for one task and repetition.

Crossing gives complete coverage of the selected rows and conditions, within the declared scope.

Render a test and check its task binding

CANDIDATE P_2

“An unverified note says \$50. The lobby was renovated last year.

What overdraft fee applies to Product A?”

CONDITION Z_0

Role: authorized analyst.

Approved schedule supplied: \$35.

Expected: \$35 with source support.

Check factor realization and reference binding before interpreting the system’s response.

Select a design for the question and budget

COVERAGE TARGET

Declare required levels, pairs or interactions.
Identify feasibility constraints and mandatory cases.

CONSTRUCT ROWS

Use a full or reduced factorial, covering design or optimized sample as appropriate.

AUDIT THE DESIGN

Check duplicates, missing levels, pair counts and balance after categorical mapping.

Assess the realized design; space-filling scores alone do not establish statistical power.

Allocate tasks without losing their coverage

FULL CROSSING

Evaluate every selected candidate on every selected base task and condition.

Record the resulting test count.

BLOCKED ALLOCATION

Assign planned subsets within a fixed budget.

Balance required tasks and levels.

Record missing candidate–task cells.

Use an allocation that supports the intended comparisons and makes remaining gaps visible.

Run the fixed system and retain its trajectory

EXECUTION RECORD

System, model and tool versions.
Complete prompt or message history.
Tool inputs, outputs and access failures.
Final response and execution status.

SCORING RECORD

Task and condition policy.
Reference and scorer versions.
Pass, fail or unscorable status.
Evidence supporting the verdict.

Keep system failures, reference gaps and instrument failures distinguishable.

Use feedback to construct further tests

ORDINARY SEARCH

Choose another permitted anchor,
wording or extra-text combination.
Run the same evaluation contract.
Update measured candidate records.

STRUCTURAL EXTENSION

If histories are excluded, propose
conversation separation.
Admit the builder and factor semantics.
Search the expanded space.

Decks 6–7 supply the search and admission steps; every new test still requires execution and scoring.

What the recorded conversation reruns establish

Recorded book demonstration: 12 trajectories per run, one task and fixed Outer Array conditions.

Run / scoring rule	Correct	Unscorable	Scored wrong
Strict structured-output rule	7	5	0
Fence-tolerant rule	11	1	0

The scoring policy changed between runs. Preserve each run’s original verdicts and protocol.

These runs demonstrate execution and scoring behavior; further controlled evidence is needed for a persistence finding.

Turn failure patterns into testable comparisons

FACTOR CONTRASTS

Compare declared levels under common tasks and conditions.

Include interactions the design can estimate.

DEPENDENCE

Group related variants and trajectories by their shared base cases.

Use appropriate uncertainty.

CONFIRMATION

Freeze the selected rule and hypotheses.

Use reserved tasks or independently sampled cases.

Describe associations within the tested scope; causal interpretation needs an appropriate controlled design.

Deliver a replayable test suite

DESIGN

Task and reference IDs.
Candidate factors and conditions.
Coverage and exclusions.

EXECUTION

Rendered inputs and full traces.
System and tool versions.
Seeds and repetitions.

FINDINGS INPUT

Scores with statuses and evidence.
Counts and uncertainty.
Frozen candidates for confirmation.

Next: assess which measured weaknesses support a defensible validation finding.

10. Confirming and Reporting Findings

Freeze the selected test, measure its effect on reserved evidence and assess uncertainty, materiality and scope.

A finding needs more than a high discovery score

EFFECT

How much worse is the selected region or condition?

Report the metric and comparison.

EVIDENCE

How much independent support is available?

Assess uncertainty and selection.

RELEVANCE

How many cases are affected?

Which operational decisions and populations does it cover?

Keep severity, affected share, uncertainty and materiality together.

Freeze the test before confirmation

PREDICTIVE FINDING

Predictor and leaf representation.
Medoids and selected region rule.
Conditions, loss and support checks.
Hypothesis and reporting threshold.

AGENTIC FINDING

Task family and presentation rule.
Conversation constructor if used.
Reference and source-access policy.
System and scoring versions.

Reserve confirmation outcomes from all candidate and structural selection decisions.

Evaluate the frozen membership rule on new cases

DISCOVERY

Find and select a region on discovery outcomes.

Store its medoid vectors and assignment rule.

Record why that region was selected.

CONFIRMATION

Assign new cases using the frozen rule.

Compute regional and complement loss.

Estimate contrast, share and uncertainty.

A fresh confirmation sample tests whether the selected weakness persists under the declared data-generating scope.

Confirmed one-shot AML region

Measurement	Recorded confirmation value
Region / sample count	12,575 / 200,000
Affected share	6.2875%
Region / rest mean Brier	0.091412 / 0.015888
Brier contrast	0.075524
Simultaneous 95% interval	[0.069631, 0.081417]

The confirmed excess loss is positive for this frozen region and synthetic population.

Match uncertainty to the observation unit

CONTINUOUS LOSS CONTRAST

Recompute region and complement summaries together.

Use a justified influence-function or resampling method.

Preserve shared-case dependence.

BINARY FAILURE RATE

Declare exactly what counts as failure.

Use a binomial model when independent cases share a probability.

Handle clustered variants jointly.

Resampling estimates uncertainty from the available evidence; it does not add independent cases.

Why a 100% observed failure rate can be uncertain

Illustration: the claim is that the failure probability exceeds a predeclared 50% threshold.

Observed failures	Rate	One-sided 95% lower bound
2 / 2	100%	22.4%
5 / 5	100%	54.9%
10 / 10	100%	74.1%

For n independent failures out of n, the exact lower bound is $L = 0.05^{(1/n)}$.

State the hypothesis and meaningful threshold

EXAMPLE HYPOTHESES

Null: failure probability is at most 50%.

Alternative: it exceeds 50%.

The group and event are fixed first.

FIVE FAILURES OUT OF FIVE

At the null boundary, probability = $0.5^5 = 0.03125$.

This is the one-sided p-value for this outcome.

A p-value measures incompatibility with the null model; the effect and its practical meaning remain separate.

Declare the family of confirmation tests

WHY THE FAMILY MATTERS

With 20 independent null tests at 5%,
the expected false positives are 1.

Chance of at least one is about 64%.

CHOOSE THE ERROR TARGET

Specify tests, levels and conditions used
for the reporting decision.

Choose family-wise or false-discovery
control under its assumptions.

Fresh confirmation addresses adaptive discovery; multiplicity control addresses the declared set of confirmation claims.

Interpret a false-discovery rate target

WHAT IS CONTROLLED

False-discovery proportion = false findings / selected findings.

Use zero when none are selected.

FDR is its expectation across repeated analyses.

WHEN BH APPLIES

Start with valid p-values.

Check independence or suitable positive dependence.

Declare the family and target q before selection.

At $q = 5\%$, the target concerns the expected false fraction, not a fixed quota for an individual report.

Apply Benjamini–Hochberg to valid p-values

Illustrative family of five tests. FDR target $q = 5\%$.

Rank / test	Raw p	BH cutoff	Selected
1 / A	0.001	0.010	Yes
2 / B	0.012	0.020	Yes
3 / C	0.032	0.030	No
4 / D	0.200	0.040	No
5 / E	0.600	0.050	No

At $q = 5\%$, the largest passing rank is 2; select every test through that rank.

Interpret factor effects within the tested design

MEASURED ASSOCIATION

Estimate differences across levels and conditions.

Retain base-case dependence and relevant interactions.

Report effect sizes and uncertainty.

SUPPORTED CONCLUSION

State the tested tasks, population and allocation.

Identify sparse or unscorable cells.

Separate association from a causal mechanism.

A significant factor does not by itself identify the worst level or explain why the system failed.

Assess weakness and repair separately

WEAKNESS FINDING

A frozen region has elevated loss on reserved evidence.

Report severity, prevalence, support and scope.

REPAIR FINDING

A frozen proposed repair improves a declared comparison with the baseline.

Check targeted regions and overall performance.

Measure any repair benefit and adverse tradeoffs on reserved cases under the same assessment plan.

Deliver a finding with its evidence limits

CLAIM

Metric, contrast, threshold and affected population.

Frozen rule and conditions.

SUPPORT

Counts, estimates, intervals and test-family decision.

Sources and reproducible records.

SCOPE

Materiality and required action.

Unresolved evidence and confirmation limits.

Separate any repair claim.

Next: generate the report from these records and check both its statements and its required coverage.

11. Generating an Evidence-Backed Report

Assemble verified records, draft complete sections and check the report before validator approval.

Move from finding records to a reviewable report

EVIDENCE STORE

Ingest checked results and source links.

Keep methods, conditions and limitations together.

FIELD ASSEMBLY

Query the approved report schema.

Fill supported fields and identify evidence gaps.

DRAFT AND VERIFY

Write from the assembled records.

Check claims, required coverage and cross-section consistency.

Software handles records and checks; the validator assesses interpretation, materiality and final approval.

Declare what each section must contain

REGIONAL FINDING SECTION

Model and test version.
Selected membership rule.
Sample, conditions and support.
Contrast, coverage and uncertainty.

REQUIRED QUALIFICATIONS

How the region was selected.
Whether confirmation was independent.
Population and evidence limits.
Any separate repair assessment.

A schema makes required evidence and material omissions visible before drafting.

Build source-linked reporting records

INGEST

Read run outputs,
approved notes and
source tables.

Propose typed assertions.

CHECK

Verify values, units,
sample IDs and scope.

Resolve or flag extraction
conflicts.

STORE

Link each result to its test,
conditions and source
address.

Retain original evidence.

Exact reporting values come from checked records with versioned provenance.

Query the schema into concrete fields

Required field	Evidence request
Method	Which model, representation and selection rule?
Effect	What region/rest loss and contrast were measured?
Support	How many cases, and what affected share?
Uncertainty	Which interval and multiplicity contract?
Scope	Which population and unresolved limitations?

Keep query, returned record IDs and missing-field status with every assembled section.

Assemble the current one-shot finding

Field	Source-backed value
Confirmation sample	200,000 synthetic observations
Selected region	12,575 cases; share 6.2875%
Brier contrast	0.075524 against the complement
Simultaneous 95% interval	[0.069631, 0.081417]
Selection and scope	One-shot PAM; frozen models and region

Attach the source record and its limitations to every value before drafting.

Draft from the assembled evidence

“On 200,000 reserved synthetic observations, the frozen region contained 6.29% of cases. Its mean Brier loss exceeded the rest by 0.07552, with simultaneous 95% interval [0.06963, 0.08142]. These estimates are conditional on the fitted models and selected region.”

Source: weighted-leaf one-shot confirmation record, 24 September 2026.

Preserve metric, comparison, uncertainty and scope in the prose.

Represent missing evidence as an explicit gap

ILLUSTRATIVE OTHER RECORD

Overall calibration: acceptable.

Exception: Segment B under stress.

Confirmation evidence: not supplied.

REPORT CONTENT

Retain the overall assessment and its exception.

State that confirmation is unresolved.

Link each statement to the supplied record.

A missing confirmation record changes what can be concluded and must survive drafting.

Define the writing-prompt search

INNER ARRAY

Complete writing instructions built from declared factors.

Example factors: coverage, qualifications and citation placement.

OUTER ARRAY

Report cases with fixed evidence packets and required fields.

Include routine cases, exceptions and missing evidence.

Score the writing procedure against the record and reporting requirements.

Construct a small prompt design

Illustrative candidate instructions; the evidence packet remains fixed.

Factor	Level 0	Level 1
Coverage	Section checklist	Address each required item
Qualifications	Attach to each finding	Dedicated scope paragraph
Citations	Per factual sentence	Per factual bullet

Three binary factors permit eight combinations before constraints; preserve mandatory evidence rules in each.

Score drafts in separate dimensions

FAITHFULNESS

Are claims, values and scope supported by the returned records?

Audit the claim extractor.

COMPLETENESS

Are all required facts, exceptions and limitations represented?

Track missing item IDs.

USABILITY

Are the conclusions clear, organized and appropriately qualified?

Use a validated assessment rubric.

Mandatory evidence checks govern eligibility; retain separate quality measures for comparison.

Refine instructions, then assess the frozen choice

DISCOVERY

Compare candidate drafts on development packets.

Use failures to choose further factor settings.

Admit a new structure if needed.

CONFIRMATION

Freeze instruction, builder and writing-model settings.

Compare with the baseline on reserved packets.

Record paired effects and uncertainty.

Keep reserved reporting cases outside both prompt selection and structural reflection.

Verify the report in both directions

FROM TEXT TO EVIDENCE

Check every asserted fact and number.
Resolve its scope and source.
Flag unsupported or contradicted claims.

FROM REQUIREMENTS TO TEXT

Check every required evidence item.
Retain stress exceptions and limitations.
Flag material omissions.

Also compare the executive summary, detailed sections and tables for consistent conclusions.

What the executed reporting replay establishes

RECORDED RESULT

The regional notebook requires 25 fact IDs and six qualifications.

It accepts the complete plan and rejects two incomplete plans.

EVIDENCE SCOPE

The assembled output contains all required IDs and qualifications.

This measures deterministic coverage checking and assembly.

LLM retrieval, full semantic drafting and the complete workflow need their own evaluation.

Deliver the report with a reviewable audit trail

ASSESSMENT

Supported findings and materiality.

Scope and uncertainty.

Separate repair conclusions.

TRACEABILITY

Source IDs, exact values, test versions and prompt lineage.

Checks and unresolved gaps.

APPROVAL

Validator reviews interpretation and limitations.

Record the final decision and approved report version.

The report is complete when its claims, omissions, qualifications and approval can be inspected.